# Technical note: searching the ASTRA quantile data vectors for the largest Fisher gain over the full-sample 2PCF

ASTRA clustering project

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**Abstract**

Using the complete grid of full-box runs on disk (nine AbacusSummit cosmologies × 50 HOD draws), we ask which ASTRA environment-split statistic, or which pair of statistics, extracts the most cosmological Fisher information beyond the plain full-sample auto-correlation. We sweep all 16 single quantile vectors and all 120 pairs, in two framings (standalone and added to the full auto), scoring each by the HOD-marginalised Fisher figure of merit. The naive ranking is dominated by vectors containing a *random*-quantile leg, with apparent gains up to 55× (all of the global top 20 carry a random leg). We test whether this is an artefact with two robustness checks — propagating the derivative noise, and refitting with a nonlinear HOD response — and find the random legs *survive both*: their environment-classified monopoles carry real cosmic-web signal, and the gains are neither a noise nor a nonlinearity artefact (only $\sigma_8$ stays untrustworthy, limited by the nine-cosmology grid). Among the *data*-leg vectors — a conservative subset — the gains are already substantial: adding a single full×data-quantile cross buys $\sim 5\times$ in the three-parameter FoM, and a greedy forward selection that spans the four density quantiles reaches a five-stem vector (`full+xdQ1+dQ1+dQ2+xdQ3+dQ4`) with $\mathrm{FoM}_3 = 43\times$ — marginalised errors tighter than the standard 2PCF by 3.4× ($\omega_b$), 4.1× ($\omega_c$), 3.6× ($n_s$) and 2.8× ($\sigma_8$), with clear diminishing returns past four stems. We document the method, the random-quadrupole diagnosis, the honest ranking, and the greedy chain.

## 1 Question and data

ASTRA splits a galaxy sample into four environment quantiles (Q1 underdense → Q4 overdense) and measures, per box, the multipoles $\xi_{0,2}(s)$ of: the full-sample auto-correlation (`tpcf_full_data`, our baseline), the four data-quantile autos (`dQ1..4`), the four random-quantile autos (`rQ1..4`), the four full×data-quantile crosses (`xdQ1..4`), and the four full×random-quantile crosses (`xrQ1..4`). That is 17 stems $\times\{\ell = 0, 2\} \times 15$ bins per run.

The signal data are the $9 \times 50 = 450$ phase-matched full-box runs (`data/fullbox/`); cosmology derivatives come from the global linear response fit (`derivative_global_*`, condition number 1.34 on the full grid). The covariance of one $500\,h^{-1}$ Mpc subbox is estimated from the mean-subtracted subboxes pooled across all nine cosmologies, $9 \times 64 = 576$ fluctuation samples, and scaled to the full $2000\,h^{-1}$ Mpc box volume ($C/64$). The HOD nuisance covariance $C_{\mathrm{HOD}}$ is the empirical scatter of the 50 same-phase `c000` draws.

## 2 Method

The analysis is implemented in `scripts/fisher_vector_search.py`, which reuses the assembly and Fisher machinery of `scripts/fisher_joint.py`. The steps are:

1. **Candidate vectors.** Each candidate is one or two ASTRA stems, taken at *native* 15-bin resolution in mono+quad ($\ell = 0, 2$). The 576-sample pooled covariance keeps the Hartlap factor benign even at $n_b = 60$ (pair) or 90 (pair added to the full auto), so no rebinning is needed. We evaluate the 16 singles and all $\binom{16}{2} = 120$ pairs.

2. **Two framings.** *Standalone* scores the candidate by itself against the full auto ("is it better than the 2PCF?"). *Additive* scores the full auto with the candidate appended ("what does ASTRA add to the 2PCF?"). The additive framing is the physical measure of marginal information and is our headline.

3. **Fisher and marginalisation.** For each candidate we build the $16 \times 16$ joint Fisher over the four cosmological parameters $\{\omega_b, \omega_c, n_s, \sigma_8\}$ and the twelve yuan23 HOD parameters, $F = D\,C^{-1}D^{\mathsf{T}}$ with Hartlap-corrected precision, add a Gaussian yuan23 prior on the HOD block, and read off the HOD-marginalised $4 \times 4$ cosmology covariance $\mathrm{Cov} = (F + F_{\mathrm{prior}})^{-1}_{[\mathrm{cosmo}]}$ (route a).

4. **Figure of merit.** We reduce Cov to

$$\mathrm{FoM}_4 = \left[\det \mathrm{Cov}_4\right]^{-1/2}, \qquad \mathrm{FoM}_3 = \left[\det \mathrm{Cov}_3\right]^{-1/2}, \tag{1}$$

   where $\mathrm{Cov}_3$ drops $\sigma_8$ (the upper-left $3 \times 3$ block, already marginalised over $\sigma_8$). The *gain* of a candidate is $\mathrm{FoM}/\mathrm{FoM}_{\mathrm{full\,auto}}$. We headline $\mathrm{FoM}_3$ because the $\sigma_8$ derivative is unreliable (Sec. 4), so a four-parameter FoM inherits that unreliability.

# 3 Background: how the covariance and the derivatives are estimated

This section explains, in simple terms, the two main ingredients of the Fisher matrix in Eq. (4): the covariance $C$ (how noisy the measurement is) and the derivatives $D$ (how the measurement changes when a parameter changes). It is written for a reader who is new to the method.

### The simulation boxes

All our data come from the AbacusSummit simulations. Each simulation is a cubic box with a side of $2000\,h^{-1}$ Mpc (about 3 billion light-years). Galaxies are placed inside the box with an HOD (halo occupation distribution) model, at redshift $z = 0.5$. We use nine different cosmologies. For each cosmology we have 50 galaxy catalogues, each one with different HOD parameters. So we have $9 \times 50 = 450$ boxes in total. All boxes use the same random "phase" (called `ph000`): they start from the same initial pattern of matter. This matters because, when we compare two boxes, the random differences between them ("cosmic variance") then mostly cancel.

### The covariance, from subboxes

The covariance tells us how much our measurement would change if we measured it again in a different patch of the Universe. Ideally we would use many separate big boxes, but big boxes are expensive to simulate. Instead we cut one full box into smaller cubes. We divide the $2000\,h^{-1}$ Mpc box into $4 \times 4 \times 4 = 64$ subboxes, each with a side of $500\,h^{-1}$ Mpc. We measure the data vector in every subbox. The scatter among the 64 subboxes gives an estimate of the covariance of one $500\,h^{-1}$ Mpc subbox.

One box gives only 64 subboxes. That is not many compared with the length of our data vector (up to 180 numbers). To do better we combine the subboxes from all nine cosmologies. For each cosmology we first subtract its own mean, so that only the random fluctuations remain, and then we put all the subboxes together. This gives $9 \times 64 = 576$ samples. Here we assume the level of noise is about the same in every cosmology, which is a good approximation over our small range of cosmologies.

Our real measurement uses the full $2000\,h^{-1}\,$Mpc box, not a small subbox. A larger volume gives a smaller error. The variance goes down in proportion to the volume, so we divide the subbox covariance by 64 (the number of subboxes in a full box) to get the covariance of the full box.

Finally, a covariance estimated from a limited number of samples gives a slightly biased result when we invert it. We correct this with the standard "Hartlap factor", $(N - n_b - 2)/(N - 1)$, where $N = 576$ is the number of samples and $n_b$ is the number of data points. For the five-stem vector ($n_b = 180$) this factor is 0.685. Using 576 samples instead of 64 is exactly what lets us keep the quadrupole at full resolution: with only 64 samples a 180-point vector could not be inverted at all (we would need $N > n_b$).

### How robust is this covariance?

It is fair to ask how much we can trust a covariance built this way. We checked it in three ways, and the short answer is: it is robust enough for *comparing* data vectors (the main goal of this note), but the *absolute* error bars should be read with some caution.

The reassuring test is a "jackknife": we rebuild the covariance nine times, each time leaving one cosmology out of the pool ($8 \times 64 = 512$ samples), and recompute the five-stem $\mathrm{FoM}_3$. The result changes by only about 6% (about 2% per parameter). So the answer does not depend on which cosmologies we pool, and because every data vector uses the *same* covariance, their ranking is trustworthy.

There are, however, three weaknesses that the jackknife does not see:

- **The subboxes are not fully independent.** The 64 subboxes come from a single box, so they share the largest waves and neighbouring subboxes are correlated. The true number of independent samples is therefore below 576. This means the real Hartlap correction should be a little stronger than the 0.685 we use, and the ratio of samples to data points (here $576/180 \approx 3.2$) is in practice even smaller. A ratio of only a few is modest: the inverse covariance is somewhat noisy, and indeed the correlation matrix has a large condition number ($\sim 10^4$), meaning some directions are close to unconstrained.

- **Open boundaries.** A $500\,h^{-1}\,$Mpc subbox has open edges and feels large-scale "super-sample" modes; the full $2000\,h^{-1}\,$Mpc box is periodic and does not. Dividing the subbox covariance by 64 assumes a simple volume scaling and ignores this difference, so the absolute covariance, especially on large scales, is only approximate.

- **Pooling assumption.** Pooling across cosmologies assumes the noise is the same for every cosmology. This is mild over our small parameter range, but it is an assumption.

In short, the *relative* results (which vector wins, by how much) are robust to roughly the 6% level; the *absolute* $\sigma$ values should be treated as estimates good to perhaps 10–20%. The clean way to remove these approximations is to estimate the covariance from many *independent* full boxes — for example the 25 AbacusSummit phases (`ph000`–`ph024`) measured as periodic boxes, or a large suite of approximate mock catalogues — which would give many more than 576 truly independent samples and drop the open-boundary problem entirely. That is the recommended next step before quoting absolute error bars.

### The derivatives, from a response model

The Fisher matrix also needs the derivatives: how the data vector changes when we change each parameter by a small amount. We get these from a "global response model". We fit one simple straight-line (linear) relation to all 450 boxes at the same time, separately for each bin of the data vector:

$$\xi \;=\; \xi_0 \;+\; \sum_p a_p\,\theta_{\mathrm{cosmo},p} \;+\; \sum_q b_q\,\theta_{\mathrm{HOD},q}, \tag{2}$$

where the $\theta_{\mathrm{cosmo}}$ are the four cosmological parameters and the $\theta_{\mathrm{HOD}}$ are the twelve HOD parameters. The slopes $a_p$ are the cosmology derivatives; the slopes $b_q$ are the HOD derivatives. Because the HOD terms are fitted at the same time, the cosmology slopes come out "clean": they are not spoiled by the fact that different cosmologies happen to have different HOD values. And because all boxes share the same phase, cosmic variance cancels in the slopes. This fit is very well behaved on the full grid (its condition number is 1.34).

### A more flexible fit: the Gaussian Process

The straight-line fit assumes the data vector changes in exact proportion to each parameter. The true response is only approximately linear. To test this we also used a Gaussian Process (GP). A GP is a flexible method that learns a smooth curve from data without assuming a straight line; as a bonus, it also gives an error bar on every prediction.

We trained one GP for each cosmology. The input was the twelve HOD parameters and the output was the data vector, using that cosmology's 50 catalogues as training points. The HOD derivative is then the slope of the GP curve at the central HOD point. Two things came out of this test. First, the HOD response is mildly nonlinear: the GP fits the data clearly better than the straight line (it captures about 75% of the variation, against about 18% for the line). Second, only a few HOD parameters — mainly the typical mass of the host halo and its scatter — drive most of the change; the other parameters matter little.

The headline numbers in this note use the simpler linear response model, because it is stable and well understood on the full grid. The GP is a refinement and a cross-check; turning it into final derivatives that also carry their own error bars is left for future work. (The same GP idea, applied separately in each cosmology and evaluated at one common HOD point, is also how we can build "HOD-matched" derivatives, but that is beyond the scope of this note.)

## 4 Are the random legs an artefact? Two robustness tests

Table 1 and the bright `rQ4`/`xrQ4`/`xrQ1` band in Fig. 4 show that the naive sweep is dominated by vectors containing a random leg ($\mathrm{FoM}_3$ up to 55×). Taken at face value this says the random-quantile statistics are the most powerful of all. Because that reverses every earlier single-parameter result, we treated it as a warning sign and ran two robustness tests. The outcome, stated up front, is that the random legs *survive* both: their gains are real signal, not an artefact.

First, the relevant physics. Once the uniform random points are classified by their Delaunay environment, the overdense quantile genuinely sits in overdense regions and the underdense one in voids, so the random-quantile *monopoles* measure the clustering of the cosmic-web environments themselves — a real, cosmology-dependent quantity, as signal-rich as the data quantiles (`rQ1`,`rQ4`

Table 1: Raw top of the additive ranking (full auto + stem(s)). A star marks a candidate that contains a random leg (`rQ` or `xrQ`). *Every* entry in the global top eight, and all 20 of the global top 20, carry a random leg, despite random legs underperforming in every earlier single-parameter test — a warning sign examined in the text.

| | vector | $\mathrm{FoM}_3$ gain | $\mathrm{FoM}_4$ gain |
|---|---|---|---|
| singles | `xrQ4`$^\star$ | 12.2 | 13.5 |
| | `rQ4`$^\star$ | 8.6 | 9.9 |
| | `xrQ1`$^\star$ | 7.8 | 8.6 |
| pairs | `xrQ1+xrQ4`$^\star$ | 55.8 | 64.6 |
| | `xdQ1+xrQ4`$^\star$ | 44.6 | 59.0 |
| | `xdQ4+xrQ4`$^\star$ | 43.4 | 60.3 |

$\ell 0$ noise/signal $\approx$ 0.05–0.09), *not* Poisson noise. The random *quadrupoles*, on the other hand, are physically weak: the data quadrupole comes from peculiar velocities, but the random points have none, so their quadrupole is dominated by realisation noise (`rQ2/3` $\ell 2$ noise/signal $\approx$ 0.7–0.8). This raises two distinct worries about the random-leg gains, which we test in turn:

**Test 1 — derivative noise (Fig. 1).** A Fisher matrix treats $D$ as exact; noise in $D$ adds $\mathrm{Tr}(C^{-1}\mathrm{Cov}(\delta)) > 0$, which it reads as constraining power. We estimate $\mathrm{Cov}(\delta)$ from the *random* part of the global-fit slope (driven by the ASTRA-iteration scatter, $\sigma^2_{\mathrm{astra}} = \overline{\mathrm{xi\_std}^2}/N_{\mathrm{iter}}$; the regression residual is not used, as it is dominated by HOD nonlinearity) and subtract the bias. The full auto, being deterministic, has zero correction (a clean check). The random legs deflate by only $\sim$25–40% and stay near the top (`xrQ4` 12.2 $\to$ 7.7, `rQ4` 8.6 $\to$ 6.0); the data legs deflate $\sim$15%. The reason is that the *global* derivative averages the per-realisation noise over all 450 phase-matched runs, so even the noise-limited quadrupoles have well-determined *global* derivatives. Derivative noise does not explain the gains.

**Test 2 — HOD nonlinearity (Fig. 2).** The global model is linear in the HOD, but the true response is mildly nonlinear, which could bias the jointly-fitted cosmology slope. We refit with HOD quadratic terms ($\theta^2_{\mathrm{HOD}}$ and $\theta_{\mathrm{HOD}}\theta_{\mathrm{HOD}}$, centred at the fiducial; design condition number 46). The cosmology derivatives move by $\sim$30–60% — but for *all* stems, with the data legs moving as much as or *more* than the random legs (data 0.32–0.60 vs random 0.24–0.48). The $\mathrm{FoM}_3$ ranking is preserved, random legs still on top. Nonlinearity therefore does not preferentially bias the random legs.

**Conclusion.** The random-leg gains are neither a noise artefact (Test 1) nor a nonlinearity artefact (Test 2); driven by their signal-rich monopoles, they carry real, usable cosmological information. Two caveats remain. *(a)* $\sigma_8$*:* the quadratic model is unstable for $\sigma_8$ ($\partial\xi/\partial \ln\sigma_8/2\xi$ goes $0.43 \to -0.15$, unphysical) — an overfit symptom of having only nine cosmologies to fix the cosmology directions, so $\sigma_8$ stays the one untrustworthy parameter and $\mathrm{FoM}_3$ (which drops it) is our score. *(b) Redundancy:* the randoms trace the same density field as the data, so their independent contribution is controlled by the data covariance (which includes the cross-correlations) — this is already in the Fisher. Given all this, the data-leg ranking in the next section is a *conservative subset* (it omits the random legs to keep the headline simple and robust); including the now-vindicated random legs would only *improve* the gains.

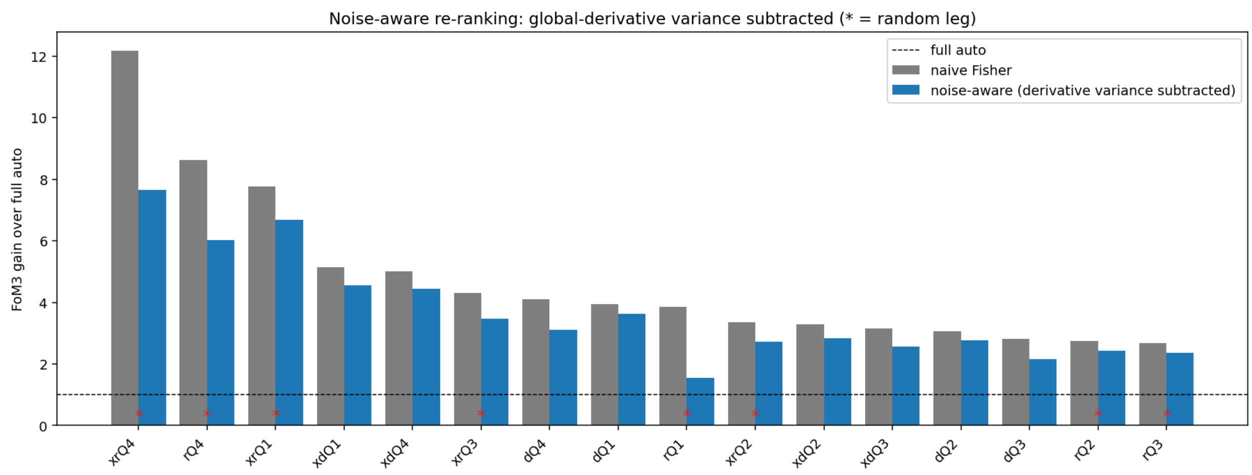


Figure 1: Test 1. Naive (grey) vs noise-aware (blue) $\mathrm{FoM}_3$ gain per single stem added to the full auto, after subtracting the derivative-noise bias $\mathrm{Tr}(C^{-1}\mathrm{Cov}(\delta))$ using the random ASTRA-iteration noise. The full auto is unchanged (zero derivative noise); the random legs ($\star$) deflate $\sim$25–40% but stay near the top — they are not a noise artefact.

Table 2: Data-leg candidates only (no random leg). Additive framing, HOD-marginalised. Baseline full-auto errors: $\sigma(\omega_b) = 5.7 \times 10^{-4}$, $\sigma(\omega_c) = 1.7 \times 10^{-3}$, $\sigma(n_s) = 1.3 \times 10^{-2}$, $\sigma(\sigma_8) = 1.4 \times 10^{-2}$.

| | vector | $\mathrm{FoM}_3$ gain | $\sigma(\omega_b)$ | $\sigma(\omega_c)$ | $\sigma(n_s)$ |
|---|---|---|---|---|---|
| singles | `xdQ1` | 5.1 | $3.3 \times 10^{-4}$ | $8.4 \times 10^{-4}$ | $8.3 \times 10^{-3}$ |
| | `xdQ4` | 5.0 | $3.2 \times 10^{-4}$ | $8.4 \times 10^{-4}$ | $8.5 \times 10^{-3}$ |
| | `dQ4` | 4.1 | $3.3 \times 10^{-4}$ | $9.8 \times 10^{-4}$ | $8.4 \times 10^{-3}$ |
| | `dQ1` | 3.9 | $4.2 \times 10^{-4}$ | $1.1 \times 10^{-3}$ | $7.7 \times 10^{-3}$ |
| pairs | `dQ1+xdQ1` | 14.0 | $2.8 \times 10^{-4}$ | — | $6.0 \times 10^{-3}$ |
| | `dQ2+xdQ1` | 12.8 | $2.5 \times 10^{-4}$ | — | $4.9 \times 10^{-3}$ |
| | `xdQ3+xdQ4` | 12.2 | $2.6 \times 10^{-4}$ | — | $5.0 \times 10^{-3}$ |

## 5 The conservative data-leg ranking

Restricting to data-leg candidates — a conservative subset, since the random legs are vindicated in Sec. 4 but omitted here for a simple, robust headline — gives a clean and physically sensible picture (Table 2, Fig. 3):

- **Singles.** The full×data-quantile crosses at the extremes lead — `xdQ1` (void) and `xdQ4` (knot), $\mathrm{FoM}_3$ gain $\approx 5$ — followed by the extreme data autos `dQ4`, `dQ1` ($\approx 4$). The middle quantiles (Q2, Q3) are weaker. Each of these improves $\omega_b$ and $n_s$ by $\sim$1.6–1.7× over the full auto. Note that on its own each vector is *worse* than the full auto (standalone $\mathrm{FoM}_3 < 1.1$); the gain is entirely degeneracy-breaking when added — the ASTRA thesis.

- **Pairs.** The best clean pair is `dQ1+xdQ1` (void data auto + void data cross), $\mathrm{FoM}_3$ gain 14, i.e. a factor $\sim$2.0 on $\omega_b$ and $\sim$2.2 on $n_s$ over the full auto. Combinations of an extreme data auto with an extreme data cross dominate; pairs of two middle quantiles, or two stems from the same family, add little (redundant).

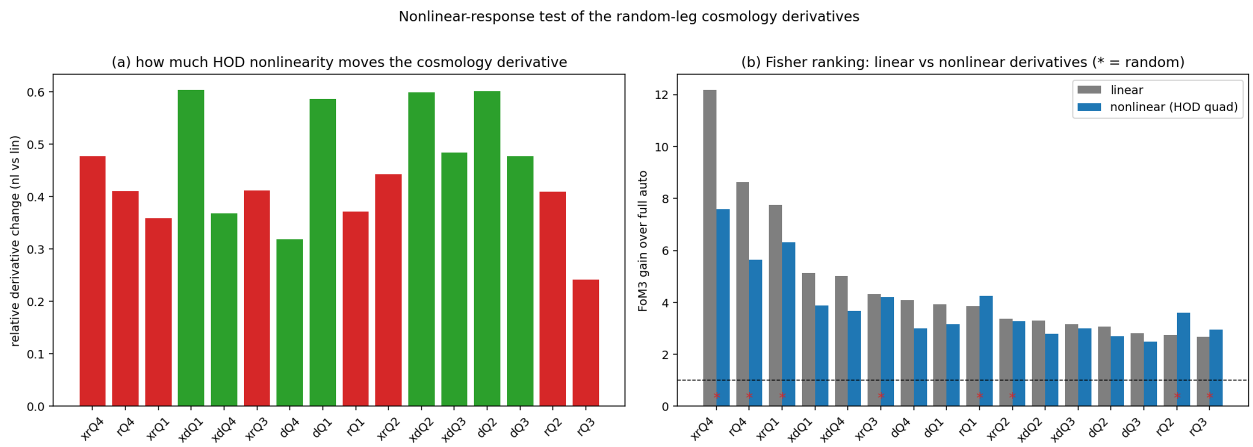


Figure 2: Test 2. (a) Relative change in the cosmology derivative when HOD nonlinearity is added to the response model: data legs (green) move as much as or more than random legs (red). (b) $\mathrm{FoM}_3$ ranking with linear vs nonlinear derivatives — the random legs ($\star$) remain on top. Nonlinearity does not single out the random legs.

Table 3: Greedy chain over the data-leg stems. "marg." is the multiplicative gain in $\mathrm{FoM}_3$ from that one step. $\sigma$ values are the HOD-marginalised errors of the cumulative vector. The chain visits all four density quantiles by step five.

| step | add | meaning | $\mathrm{FoM}_3$ | marg. | $\sigma(\omega_b)$ | $\sigma(n_s)$ | $\sigma(\sigma_8)$ |
|---|---|---|---|---|---|---|---|
| 0 | — | full auto | 1.0 | — | $5.7\times10^{-4}$ | $1.3\times10^{-2}$ | $1.4\times10^{-2}$ |
| 1 | `xdQ1` | void cross | 5.1 | 5.1 | $3.3\times10^{-4}$ | $8.3\times10^{-3}$ | $1.0\times10^{-2}$ |
| 2 | `dQ1` | void auto | 14.0 | 2.7 | $2.8\times10^{-4}$ | $6.0\times10^{-3}$ | $7.5\times10^{-3}$ |
| 3 | `dQ2` | low–mid auto | 24.7 | 1.8 | $2.3\times10^{-4}$ | $4.3\times10^{-3}$ | $6.8\times10^{-3}$ |
| 4 | `xdQ3` | mid–high cross | 32.5 | 1.3 | $2.1\times10^{-4}$ | $3.8\times10^{-3}$ | $5.4\times10^{-3}$ |
| 5 | `dQ4` | knot auto | 43.3 | 1.3 | $1.7\times10^{-4}$ | $3.6\times10^{-3}$ | $5.0\times10^{-3}$ |

- **$\sigma_8$ is untouched.** No vector improves $\sigma_8$ appreciably ($1.4\times10^{-2}\to\sim1.0\times10^{-2}$), consistent with $\sigma_8$ being amplitude-degenerate with the HOD and with its derivative being unreliable. This is why $\mathrm{FoM}_3$ (which drops $\sigma_8$) is the trustworthy score; Fig. 5 confirms the per-parameter picture.

# 6 Greedy forward selection: how many stems, and which?

The single- and pair-level results raise the obvious question: if one extreme data cross already buys $\sim5\times$ and the void auto+cross pair buys $14\times$, how far does it go? We answer it with a *greedy forward selection* restricted to the eight data-leg stems (`scripts/fisher_greedy_chain.py`). Starting from the full auto, at each step we append the single data-leg stem that most increases $\mathrm{FoM}_3$, keep it, and repeat, up to five added stems. Greedy selection is not guaranteed to find the globally optimal subset, but it is the natural way to read off the *marginal* value of each successive statistic and to expose redundancy.

The chain is `full+xdQ1+dQ1+dQ2+xdQ3+dQ4` (Table 3, Fig. 6). Three things stand out.

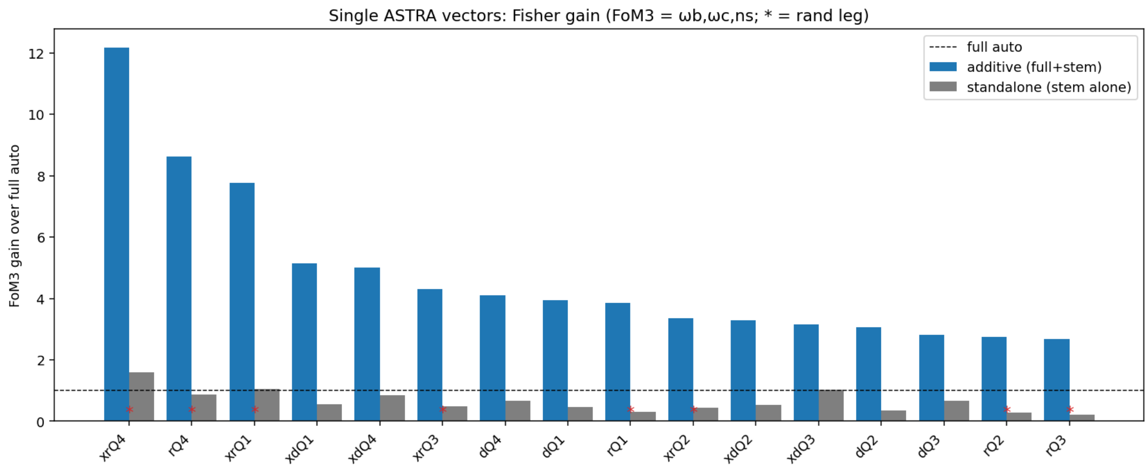


Figure 3: Single-vector $\mathrm{FoM}_3$ gain over the full auto, additive (blue) and standalone (grey), ranked. Red stars mark random-leg vectors; these are tested and found robust (not artefacts) in Sec. 4. Every vector is at or below the full-auto line on its own; the gain is degeneracy-breaking, realised only when appended.

- **Strong diminishing returns.** The per-step gain falls $5.1 \to 2.7 \to 1.8 \to 1.3 \to 1.3$. The first two stems carry most of the information ($\mathrm{FoM}_3$ reaches 14 of the eventual 43 by step two); beyond step four each new statistic adds only $\sim 30\%$, while the bin count and Hartlap/redundancy cost keep rising. A four-stem vector is the practical sweet spot.
- **Gains come from spanning environments, not repetition.** Greedy does not pile up void statistics; after the void auto+cross it reaches for a *different* density bin (`dQ2`), then mid–high (`xdQ3`), then the knot (`dQ4`), touching all four quantiles by step five. Different environments respond to cosmology and to the HOD in different ways, so each new environment breaks a *complementary* degeneracy; a second statistic from the same environment would be largely redundant.
- **$\sigma_8$ moves only in combination.** No single vector improved $\sigma_8$ (Sec. 5), but the cumulative five-stem vector tightens it $2.8\times$ ($1.4\times10^{-2} \to 5.0\times10^{-3}$). The multi-environment combination partially lifts the amplitude degeneracy that defeats any one statistic — though $\sigma_8$ remains the least trustworthy entry because its derivative is unreliable, and $\mathrm{FoM}_3$ (which excludes it) stays our headline score.

## The five-stem vector versus the 2PCF

Figure 7 is the head-to-head Fisher comparison of the standard full-sample 2PCF (`tpcf_full_data`, mono+quad) and the five-stem ASTRA vector, both HOD-marginalised, generated by `scripts/fisher_compare_fu` The five-stem 68/95% contours sit nested well inside the full-auto contours on every panel, with the marginalised errors tighter by

$$\sigma(\omega_b)\ 3.4\times, \quad \sigma(\omega_c)\ 4.1\times, \quad \sigma(n_s)\ 3.6\times, \quad \sigma(\sigma_8)\ 2.8\times, \tag{3}$$

i.e. $\mathrm{FoM}_3 = 43\times$. Crucially the blue ellipses are not merely smaller but *rotated* relative to grey: the environment splits change the orientation of the parameter degeneracies, which is the geometric

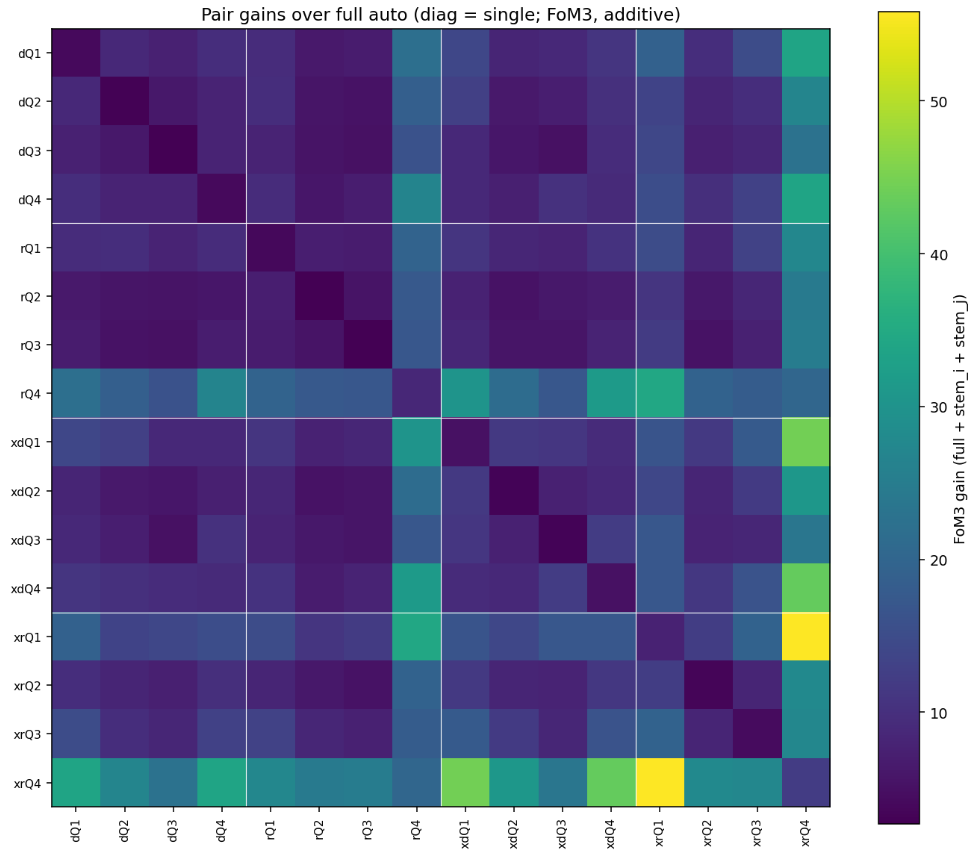


Figure 4: Additive $\mathrm{FoM}_3$ gain for every pair, full auto + $\mathrm{stem}_i$ + $\mathrm{stem}_j$ (diagonal = single). White lines separate the four families (`dQ`, `rQ`, `xdQ`, `xrQ`). The bright band on the `rQ4`/`xrQ4`/`xrQ1` rows and columns is from the random legs, which survive the robustness tests of Sec. 4; the data-leg quadrants (`dQ`, `xdQ`) are the conservative subset used for the headline.

signature of genuinely new information rather than a uniform rescaling. The five-stem 95% region is inside the full-auto 68% region for most parameter pairs — a qualitatively different constraint from a single $2\,h^{-1}\,$Mpc box, by reanalysis alone.

## 7 Anatomy of the 5-stem Fisher computation

This section opens up the Fisher calculation behind Fig. 7 step by step, so every ingredient is visible. The five-stem vector concatenates six pieces (the full auto plus the five selected stems), each in mono+quad at the native 15-bin resolution, for a data vector of $n_b = 6 \times 2 \times 15 = 180$ bins. All figures are produced by `scripts/fisher_5stem_details.py`.

The Fisher matrix is

$$F \;=\; D\, C^{-1}\, D^{\mathsf{T}}, \tag{4}$$

where $D$ is the $(4 + 12) \times n_b$ matrix of derivatives of the data vector with respect to the four cosmological and twelve HOD parameters, and $C$ is the data covariance. The cosmology constraints follow from inverting either the cosmology block alone (conditional, HOD fixed) or the full matrix plus the HOD prior and taking the cosmology sub-block (marginalised). The four steps below are the four factors in Eq. (4).

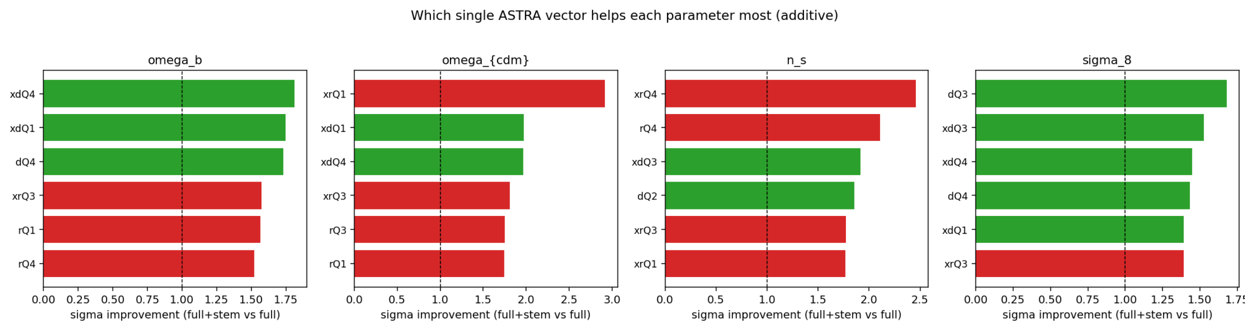


Figure 5: For each cosmological parameter, the single ASTRA vectors that most improve its marginalised error when added to the full auto. Green bars are data-leg vectors; red bars are random-leg vectors, found robust to the derivative-noise and nonlinearity tests of Sec. 4.

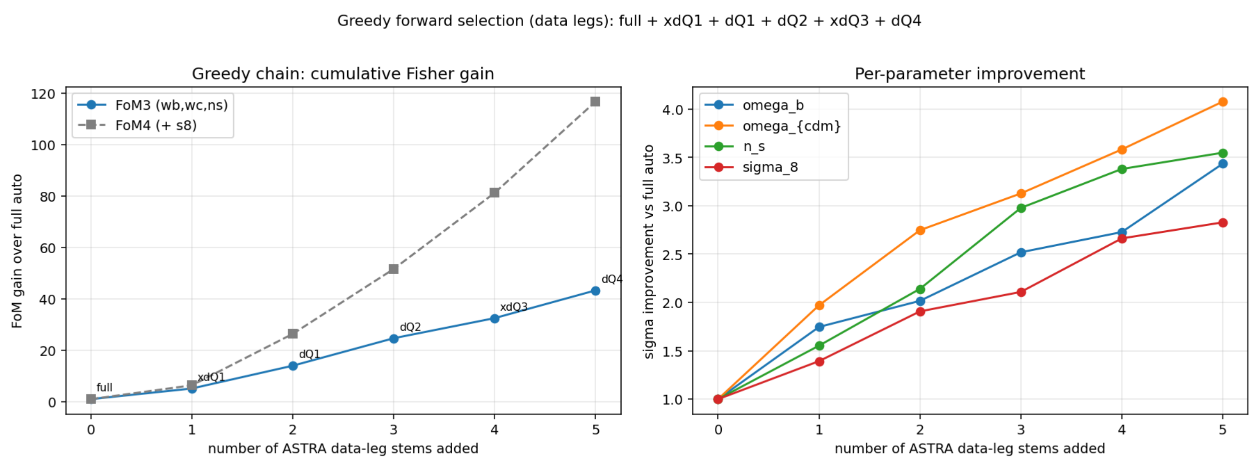


Figure 6: Greedy forward selection over the data legs. Left: cumulative $\mathrm{FoM}_3$ (and $\mathrm{FoM}_4$) gain over the full auto versus the number of ASTRA stems added — the curve flattens after $\sim$4 stems. Right: per-parameter $\sigma$ improvement over the full auto. Labels mark the stem chosen at each step.

## Step 1 — the data vector

Figure 8 shows the fiducial (`c000` ensemble-mean) data vector, one panel per piece. The monopole amplitude grows monotonically from the void (`dQ1`) to the knot (`dQ4`, an order of magnitude larger): the overdense quantile is the most strongly clustered and highest-bias tracer, the underdense one the least. The quadrupoles ($\ell = 2$) carry the redshift-space anisotropy. These six curves, stacked $[\,\ell 0, \ell 2\,]$ per piece, are the 180-component vector whose information content we are dissecting.

## Step 2 — the derivatives $D$

Figure 9 shows the four cosmology rows of $D$ (the global-response derivatives $\partial\xi/\partial\theta$) across all 180 bins, with the six piece blocks marked. Two features matter for the Fisher information:

- The derivative amplitude is largest on the high-bias `dQ4` piece, especially at small $s$ — the knot quantile is the most cosmology-responsive.
- The *shape* of the response differs from piece to piece (e.g. the $\omega_b$ acoustic feature versus the broadband $n_s$ tilt appear with different relative weights in each environment). It is precisely

this piece-to-piece difference in $\partial\xi/\partial\theta$ that lets the environment splits break parameter degeneracies the full auto cannot.

(The twelve HOD rows of $D$, not shown, enter the marginalisation in Step 4.)

### Step 3 — the covariance $C$

Figure 10 is the $180 \times 180$ correlation matrix of the data vector, estimated from the mean-subtracted subboxes pooled across all nine cosmologies (576 samples) and scaled to the full-box volume. The block structure is the crux of how much *independent* information the vector holds:

- Within each piece, neighbouring $s$-bins are positively correlated, and the monopole and quadrupole blocks are cross-correlated.
- Across pieces, the full auto is strongly correlated with the full×data crosses (`xdQ1`,`xdQ3`) because they literally share the full-sample leg, and the quantile autos are mutually correlated through the shared density field. These off-diagonal correlations are why the gains saturate: the pieces are not independent, and $C^{-1}$ correctly down-weights the redundancy.

The 576 pooled samples are what make $n_b = 180$ affordable: the Hartlap factor is $(576 - 180 - 2)/(576 - 1) = 0.685$, comfortably positive, where the 64-subbox estimate would have failed ($n_b > 64$).

### Step 4 — the Fisher matrix and parameter covariance

Combining $D$ and $C$ via Eq. (4) with the Hartlap-corrected precision gives the parameter covariance. Figure 11 shows the $4 \times 4$ cosmology parameter correlation matrices, conditional (HOD fixed) and marginalised (HOD nuisances marginalised with the yuan23 prior), with the per-parameter $\sigma$ in the titles:

| | $\sigma(\omega_b)$ | $\sigma(\omega_c)$ | $\sigma(n_s)$ | $\sigma(\sigma_8)$ |
|---|---|---|---|---|
| conditional (HOD fixed) | $2.9 \times 10^{-5}$ | $2.0 \times 10^{-4}$ | $7.6 \times 10^{-4}$ | $2.9 \times 10^{-3}$ |
| marginalised | $1.7 \times 10^{-4}$ | $4.1 \times 10^{-4}$ | $3.6 \times 10^{-3}$ | $5.0 \times 10^{-3}$ |

Two lessons:

- **Marginalising over the HOD costs a factor few** per parameter ($6\times$ on $\omega_b$, $5\times$ on $n_s$): the cosmology directions that overlap the HOD response are degraded most. This is the honest price of not knowing the galaxy–halo connection, and it is the regime all our headline numbers are quoted in.
- **Marginalisation re-shapes the degeneracies.** Conditionally, $\omega_b$ is strongly anti-correlated with $\sigma_8$ ($-0.84$) and $\omega_c$ with $n_s$ ($-0.79$); after HOD marginalisation these soften and new correlations appear ($\omega_b$–$\omega_c = +0.55$, $n_s$–$\sigma_8 = -0.42$). The rotation of the ellipses in Fig. 7 is exactly this change of correlation structure.

Table 4: Multipole configuration comparison, HOD-marginalised. $\text{FoM}_3$ gain is relative to the standard 2PCF (full auto, mono+quad). The quadrupole is weak alone but adds substantially when combined, and is the main carrier of $\sigma_8$ information.

| vector | config | $n_b$ | $\text{FoM}_3$ gain | $\sigma(\sigma_8)$ |
|---|---|---|---|---|
| full auto | mono ($\ell 0$) | 15 | 0.41 | $3.6 \times 10^{-2}$ |
| full auto | quad ($\ell 2$) | 15 | 0.12 | $4.0 \times 10^{-2}$ |
| full auto | mono+quad | 30 | 1.00 | $1.4 \times 10^{-2}$ |
| 5-stem | mono ($\ell 0$) | 90 | 12.1 | $1.2 \times 10^{-2}$ |
| 5-stem | quad ($\ell 2$) | 90 | 2.6 | $1.0 \times 10^{-2}$ |
| 5-stem | mono+quad | 180 | 43.3 | $5.0 \times 10^{-3}$ |

# 8 Monopole, quadrupole, or both?

The vectors above use both multipoles; does the quadrupole earn its place? We test this two ways with `scripts/fisher_multipole_compare.py`: a configuration comparison (mono-only, quad-only, mono+quad) for both the full auto and the five-stem vector, and a greedy "poll" that treats each (piece, multipole) as one of twelve selectable units and records whether the monopole or quadrupole is chosen at each step (Fig. 12).

- **The monopole is the workhorse.** On its own it carries most of the information — 0.41 of 1.0 for the full auto, 12 of 43 for the five-stem vector. The quadrupole alone is weak (0.12 and 2.6).
- **But the quadrupole is not redundant.** Adding it to the monopole multiplies $\text{FoM}_3$ by 2.4× for the full auto (0.41 → 1.0) and 3.6× for the five-stem vector (12 → 43). Crucially it is the main source of $\sigma_8$ information: mono-only $\sigma(\sigma_8)$ improves 2.4× when the quadrupole is added ($1.2 \times 10^{-2} \to 5.0 \times 10^{-3}$ for the five-stem vector) — the redshift-space growth signal the isotropic monopole cannot see.
- **The quadrupole matters more for ASTRA than for the 2PCF** (3.6× vs 2.4×): environment-dependent velocity fields give the quantile quadrupoles extra independent RSD information.
- **The greedy poll confirms the hierarchy.** Of the first six selected (piece, multipole) units, four are monopoles and two are quadrupoles; greedy front-loads the information-dense monopoles but reaches a quantile quadrupole (`dQ2`-$\ell 2$) as early as step four, so certain quadrupoles carry strong *independent* information.

The verdict is to use mono+quad: the monopole is primary, the quadrupole is complementary (mostly $\sigma_8$ and $n_s$), and its value is larger in the environment splits than in the plain 2PCF. This refines the earlier impression that "the quadrupole adds almost nothing" — that held for $\omega_b$ specifically and under a Hartlap-limited bin budget; with the pooled covariance the quadrupole clearly pays off.

# 9 Conclusions

1. The systematic sweep of all 16 singles and 120 pairs is dominated, at face value, by vectors with a random-quantile leg (gains up to 55×). We treated this as a warning sign and tested it

two ways: subtracting the derivative-noise bias (the random legs deflate only $\sim$25–40% and stay on top), and refitting with a nonlinear HOD response (the cosmology derivatives move $\sim$30–60% for *all* stems, data legs as much as random). The random legs survive both — their environment-classified *monopoles* carry real cosmic-web signal, so the gains are neither a noise nor a nonlinearity artefact (Sec. 4). The one direction that stays untrustworthy is $\sigma_8$, limited by the nine-cosmology grid.

2. Among the **data legs** — a conservative subset that omits the (now-vindicated) random legs for a simple, robust headline — the full×data-quantile crosses at the density extremes (`xdQ1`, `xdQ4`) are the best single additions ($\mathrm{FoM}_3$ gain $\approx 5$), and the best clean pair is the void data auto with the void data cross, `dQ1+xdQ1` ($\mathrm{FoM}_3$ gain 14, $\sim$2.4× per parameter on $\omega_b, \omega_c, n_s$). The gains are pure degeneracy breaking: each vector is worse than the full auto alone.

3. A **greedy chain** over the data legs (`full+xdQ1+dQ1+dQ2+xdQ3+dQ4`) reaches $\mathrm{FoM}_3 = 43\times$ at five stems, tightening the standard-2PCF errors by 3.4–4.1× on $\omega_b, \omega_c, n_s$ and 2.8× on $\sigma_8$ (Fig. 7). Returns diminish sharply ($5.1 \to 2.7 \to 1.8 \to 1.3$ per step), so $\sim 4$ stems is the practical optimum, and the gains come from *spanning* the four environments rather than repeating one.

4. Single vectors do not improve $\sigma_8$ (HOD-amplitude degeneracy, unreliable derivative), but the multi-environment combination does so modestly; $\mathrm{FoM}_3$ excluding $\sigma_8$ remains the trustworthy score.

5. **Use both multipoles.** The monopole is the workhorse and the quadrupole is weak alone, but adding the quadrupole multiplies $\mathrm{FoM}_3$ by 2.4–3.6× and is the dominant source of $\sigma_8$ information; its value is larger for the ASTRA splits than for the plain 2PCF (Sec. 8).

6. **Next steps.** (i) Fold the now-vindicated random legs back into the greedy chain and the optimal data vector (their signal-rich monopoles should add beyond the data legs); (ii) settle $\sigma_8$ — the only untrustworthy direction — which the nine-cosmology grid cannot pin down and which needs more cosmologies, not reanalysis; (iii) estimate the covariance from many independent full boxes (e.g. the 25 AbacusSummit phases) before quoting absolute error bars, since the relative rankings are robust to the $\sim$6% level (Sec. 3) but the absolute $\sigma$ rely on the subbox approximation.

All numbers are reproduced by `scripts/fisher_vector_search.py` (the full sweep; ranked table in `plots/vector_search/vector_search_results.csv`), `scripts/fisher_greedy_chain.py` (the greedy chain), and `scripts/fisher_compare_full_vs_5stem.py` (the corner comparison), and `scripts/fisher_5stem_details.py` (the step-by-step anatomy of Sec. 7), and `scripts/fisher_multipole_compar` (the monopole/quadrupole study of Sec. 8).

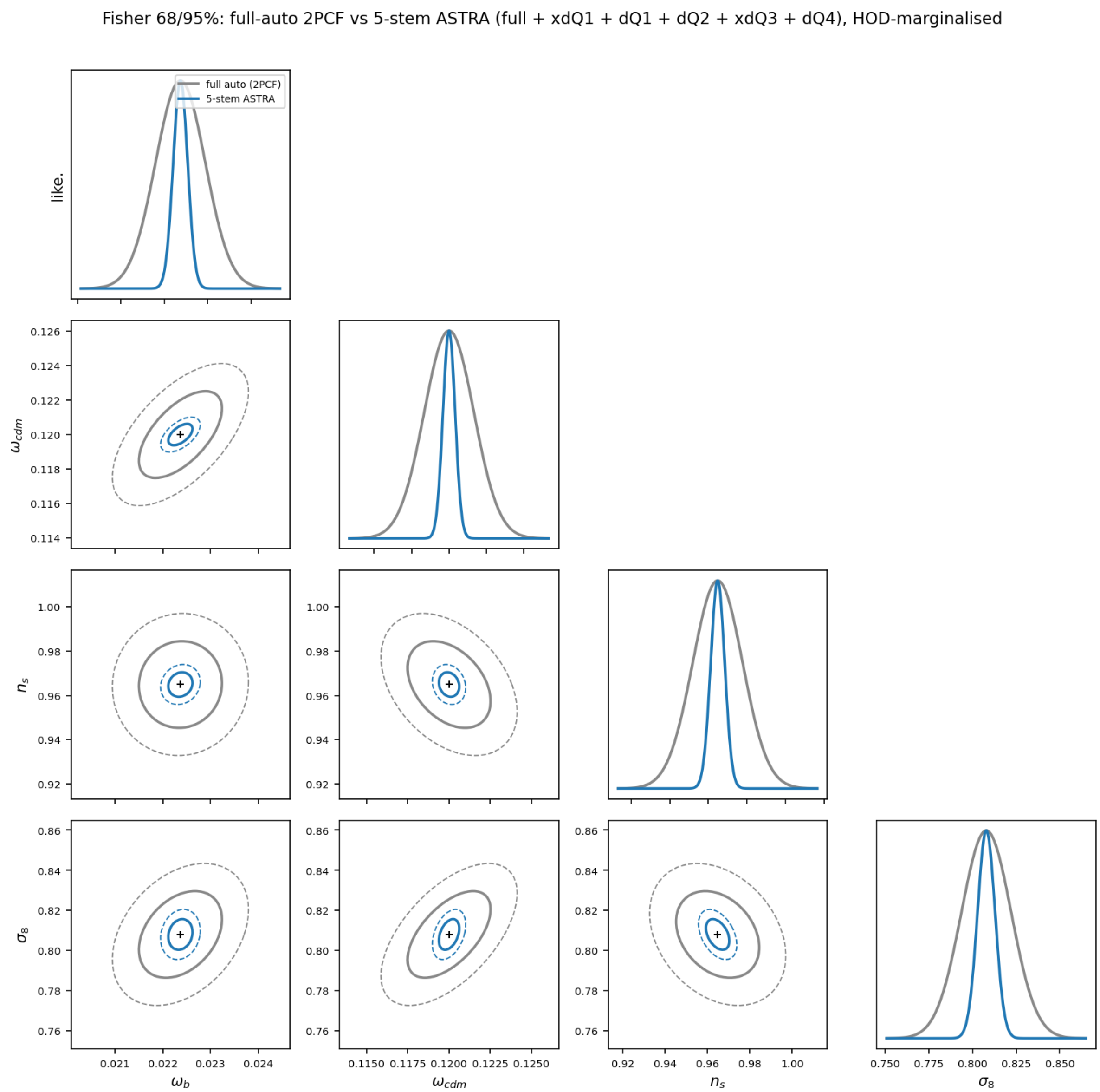


Figure 7: HOD-marginalised Fisher constraints: full-auto 2PCF (grey) versus the five-stem ASTRA vector `full+xdQ1+dQ1+dQ2+xdQ3+dQ4` (blue), 68% (solid) and 95% (dashed). The blue contours are both smaller and rotated — ASTRA shrinks and re-orients the cosmology degeneracies.

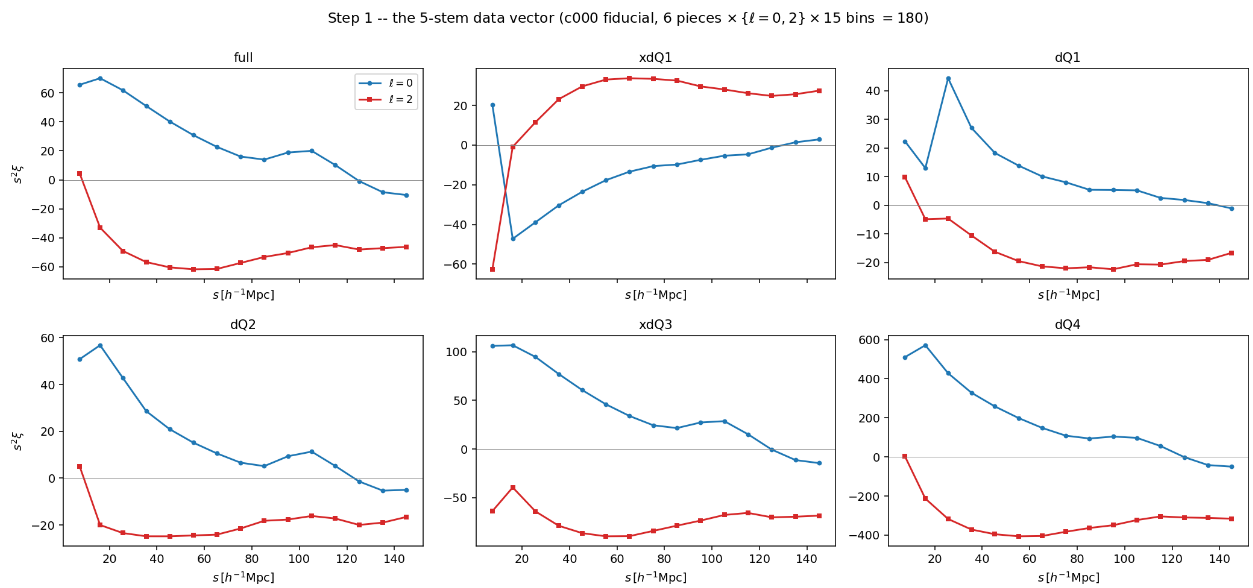


Figure 8: Step 1: the fiducial five-stem data vector, $s^2\xi_{0,2}(s)$ per piece. Amplitude grows void (`dQ1`) $\rightarrow$ knot (`dQ4`).

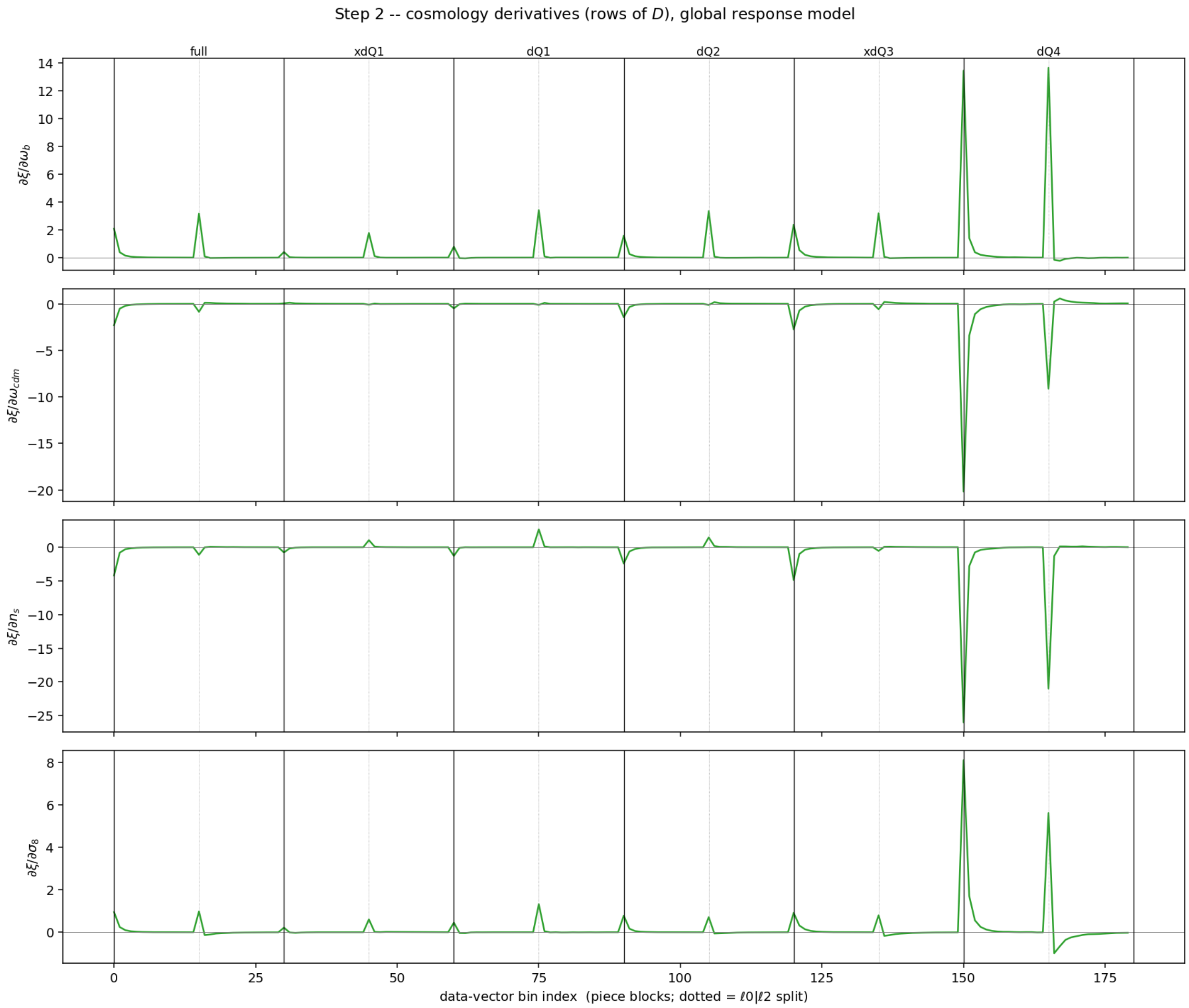


Figure 9: Step 2: the four cosmology rows of $D$, $\partial\xi/\partial\theta$, across the 180-bin vector. Solid lines separate the six pieces; dotted lines mark the $\ell 0|\ell 2$ split within each piece. The high-bias `dQ4` piece carries the largest response.

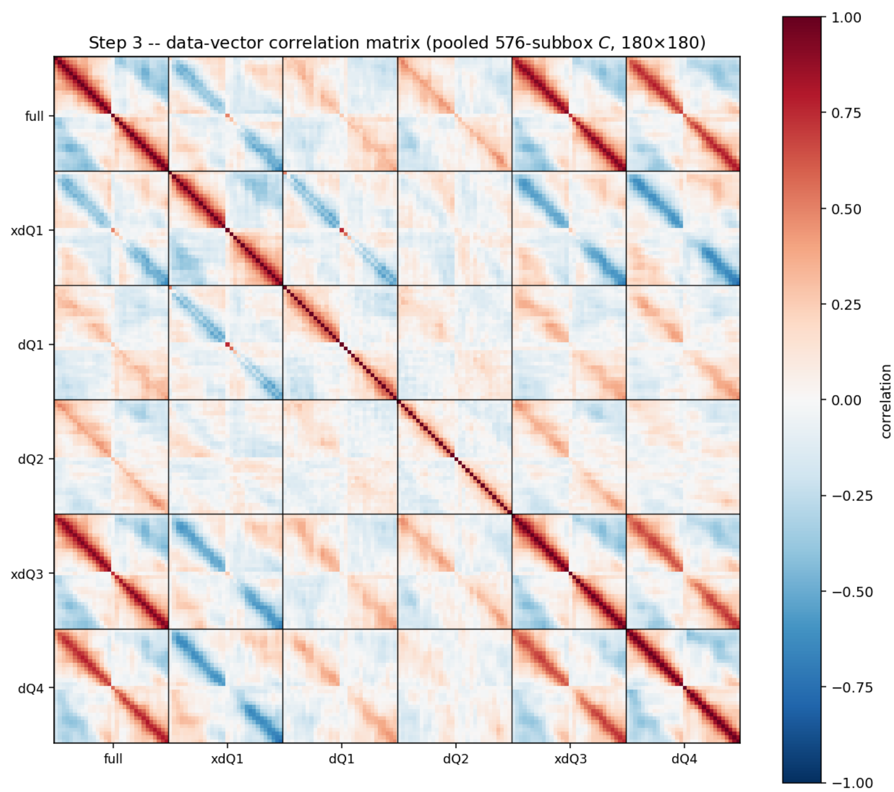


Figure 10: Step 3: the $180 \times 180$ data-vector correlation matrix (pooled 576-subbox $C$). Black lines separate the six pieces. Strong cross-piece correlations (e.g. full with the `xd` crosses) are the redundancy that limits the cumulative gain.

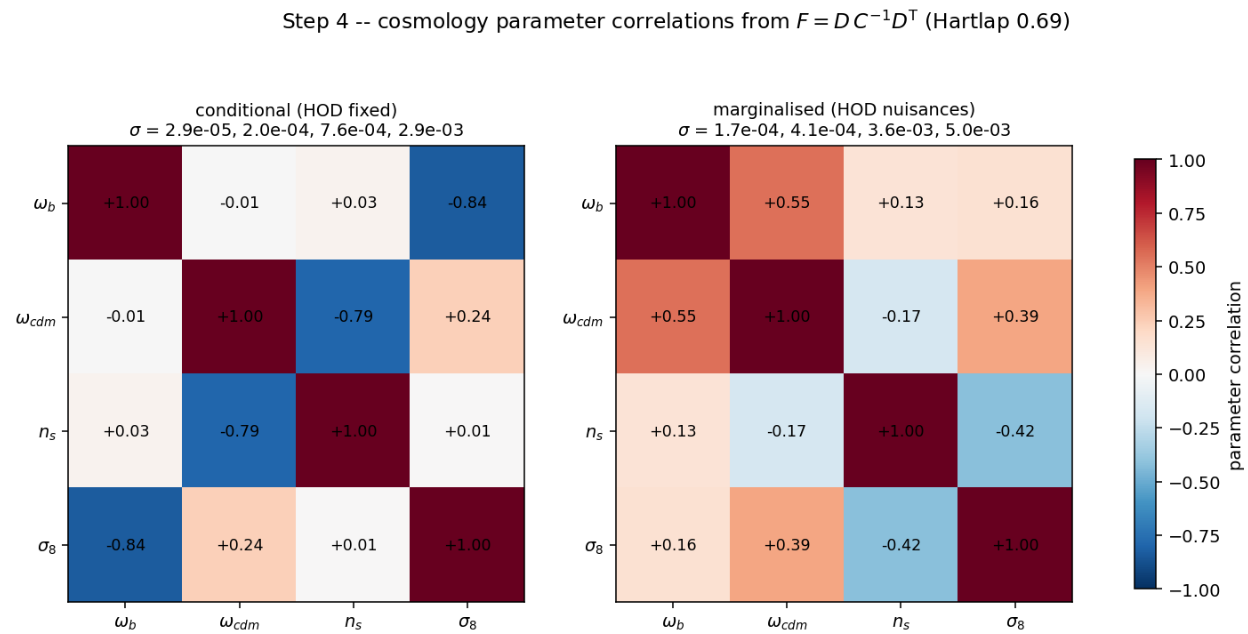


Figure 11: Step 4: cosmology parameter correlation matrices from $F = D\,C^{-1}D^{\mathsf{T}}$, conditional (left) and HOD-marginalised (right), with the per-parameter $\sigma$ in the titles. HOD marginalisation both inflates the errors and re-shapes the degeneracies.

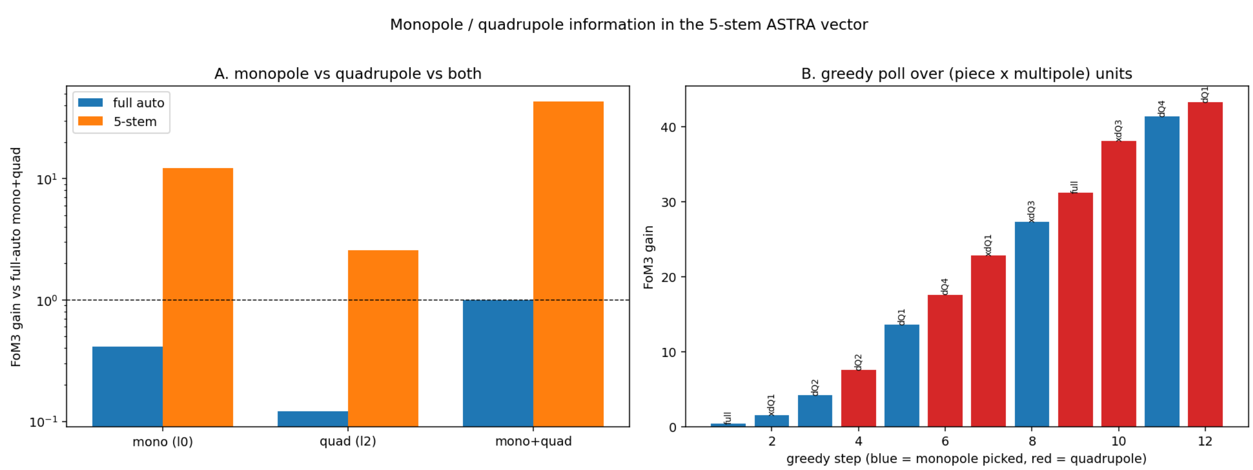


Figure 12: Left: $\mathrm{FoM}_3$ gain (log scale) for mono-only, quad-only and mono+quad, for the full auto and the five-stem vector. Right: greedy poll over the twelve (piece, multipole) units; blue bars are monopole picks, red are quadrupole picks, with the cumulative $\mathrm{FoM}_3$ at each step.